



Testable Questions



I can...

- ✓ **form a testable question about an observed phenomenon that can be answered using the Scientific Method.**
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Vocabulary

a question that can be answered with measurements in an experiment



testable question

Think & Talk

1. How does being specific make a question more testable?
2. Why can't opinions or beliefs be tested scientifically?
3. When testing a question, how many **variables** can you change at one time?

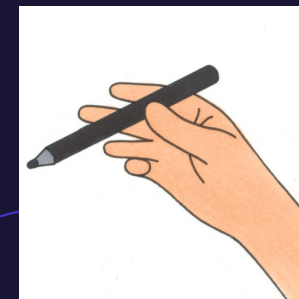
Vocabulary

Think & Talk

the factor that can
change in an experiment



1. Why can't you change more than one factor in an experiment?
2. Can you think of an example experiment that shows why only one variable can change?



Testable Questions have two parts:

- An independent variable
- A dependent variable



3 Kinds of variables

INDEPENDENT VARIABLE



What I CHANGE



DEPENDENT VARIABLE

What I OBSERVE



CONTROLLED VARIABLE

What I KEEP THE SAME

Example of a Testable Question

How does the amount of time spent studying affect test scores?

What am I testing? (independent variable)

- Amount of time spent studying

How will I measure what I am testing?

- The test scores (dependent variable)

How to write a testable question

- How does changing (independent variable) affect (dependent variable)
- What is the effect of (independent variable) on (dependent variable)
- Which (independent variable) is/ has (dependent variable) ?

Remember to ask yourself:

What am I testing? (independent variable)

How will I measure what I am testing? (dependent variable)

In your science notebook, make your own testable questions using these words:

1. Sunlight , Plant growth
2. Brand of dishwashing liquid, bubbles

Asking better questions

- **Testable and Non-Testable questions often have similar wording**
- **Non testable questions can be improved to become testable questions.**

How does a paper airplane fly?

Is this question testable? Why or why not?

Let's Make the Question Testable!

- Decide what you are going to change (independent variable)

For example:

- Size of the plane
- Style of the plane
- Type of paper



- Decide what you are going to measure (dependent variable)

For example:

- Distance the plane will fly
- Length of time the plane stays in the air



Make the Question Testable

- Now select one independent variable and one dependent variable
- Put them together to create the testable question:

"How does the type of paper affect the distance a paper airplane flies?"



Copy the T-Chart shown. Write the number of each question in the proper column and why not for the non-testable questions.

Testable Questions	Non-testable Questions And Why
1. Do pet birds like sunflower seeds?	5. Who is the best athlete in America?
2. Do bandaged cuts heal faster than uncovered cuts?	6. Is pizza healthy for kids?
3. Do dishwashers kill more germs on dishes than hand washing and air drying?	7. How does the sun burn your eyes?
4. Will a plant grow better when given rainwater or water from the city supply?	8. Which surface makes a ball roll fastest, carpet or concrete?
	9. Which bounces higher: soccer ball or basketball?

Bubble Gum

A classmate asks you what the best brand of bubble gum is. You decide to use science to find out. Create a testable question that can help you reach a conclusion.